

Grade 3 Ontario Curriculum Algebra

Balanced Equations

Unit Overview: Balanced Equations

Grade: Grade 3 | **Strand:** Algebra

Target Expectations: C2.1, C2.2, C2.3

Learning Goal: Children will describe and use variables; determine whether expressions across all four operations are equivalent; and find equivalent relationships for whole numbers up to 1000.

Lesson	Lesson Title	Learning Goal
Day 1	Variables in Real Life	I can describe how variables are used in real-life contexts.
Day 2	Equivalent in Addition and Subtraction	I can determine whether addition and subtraction expressions are equivalent.
Day 3	Equivalent in Multiplication and Division	I can determine whether multiplication and division expressions are equivalent.
Day 4	Many Ways to Make 100, 250, 500	I can find many equivalent expressions for whole numbers up to 1000.
Day 5	Balance Puzzle Capstone	I can create and solve a balance puzzle using a variable.

Lesson 1: Variables in Real Life

Goal: A variable stands for an unknown OR a quantity that varies.

Expectation: C2.1

Learning Goal: "I can describe how variables are used in real-life contexts."

Materials: M1_variable_chart, M9_equation_chart, M10_engineer_poster

1. Minds On (12 mins): Eddy's Variable Tour

Eddy: Variables are everywhere — temperature, score, age. They CHANGE every day. Eddy opens: 'Letters in math STAND for NUMBERS!'

Bring children to the carpet. Eddy puppet. 'Friends — I'm Eddy the Equation Engineer. Today I want you to see VARIABLES everywhere in your life.'

2. Action (25 mins): Variable Real-Life Stations

- Whole group: identify variables in 4 real-life contexts.
- Three stations (~7 min each).
- Stretch challenge at Station C: write a variable for a quantity in your home (steps to school, homework time).
- Discuss the two uses of variables.
- Circulate to record formative evidence on the variable tracker.
- STRATEGY COMPARE: Two pairs share their variable choices.

3. Consolidation (13 mins): Eddy's Variable Brag

Discuss:

- What does VARIABLE mean?
- What's a variable in your day?
- Difference between unknown and varying variables?
- Why do we use letters as variables?
- Eddy's preview: tomorrow we put expressions on Vex's balance scale (Lesson 2 — C2.2).

Exit: Eddy's variable refrain — class chants 3x: 'Letters in math STAND for NUMBERS!' (with hand gesture: open palm = unknown).

Lesson 2: Equivalent in Addition and Subtraction

Goal: Two addition or subtraction expressions are EQUIVALENT when they have the SAME total.

Expectation: C2.2

Learning Goal: "I can determine whether addition and subtraction expressions are equivalent."

Materials: M2_balance_mat, M3_addition_cards, M7_two_colour_counters, M8_number_line

1. Minds On (12 mins): Eddy and Vex Diagnose Equations

Eddy: $50 + 25 = 75$. Vex: $100 - 25 = 75$. Are these equivalent? Eddy opens: 'COMPUTE BOTH, then COMPARE!' Yesterday's variables are now on the balance scale.

Bring children to the carpet. Eddy + Vex puppets. 'Friends — Eddy and Vex disagree. Eddy says $50 + 25$ makes 75. Vex says $100 - 25$ makes 75. Who's right?'

2. Action (25 mins): Add/Sub Equivalence Stations

- Whole group: test 4 pairs together.
- Three stations (~7 min each).
- Stretch challenge at Station C: write a 3-term equivalent pair ($50+25+25$ vs 100).
- Discuss patterns.
- Circulate to record formative evidence on the add/sub tracker.
- STRATEGY COMPARE: Two pairs share the same equivalent pair ($50+25 = 100-25$).

3. Consolidation (13 mins): Eddy's Equivalence Brag

Discuss:

- What does EQUIVALENT mean?
- Can a + and a - expression be equivalent?
- What's compensation?
- How do you check on the balance?

Exit: Eddy's balance refrain — class chants: 'Eddy's rule — COMPUTE BOTH, then COMPARE.'

Lesson 3: Equivalent in Multiplication and Division

Goal: Two expressions are EQUIVALENT when their products or quotients are the same value.

Expectation: C2.2

Learning Goal: "I can determine whether multiplication and division expressions are equivalent."

Materials: M2_balance_mat, M4_multiplication_cards, M7_two_colour_counters, M8_number_line

1. Minds On (12 mins): Eddy Tests Mult/Div Pairs

Eddy: $4 \times 6 = 24$. Vex: $8 \times 3 = 24$. Are these equivalent?

Bring children to the carpet. Eddy + Vex puppets. 'Friends — yesterday we tested + and -. Today we test $\times$ and $\div$.'

2. Action (25 mins): Mult/Div Equivalence Stations

- Whole group: test 4 pairs.
- Three stations.
- Stretch challenge at Station C: factor a number into 3+ equivalent multiplications.
- Discuss patterns.
- Circulate to record formative evidence on the mult/div tracker.

3. Consolidation (13 mins): Eddy's Mult/Div Brag

Discuss:

- Can a multiplication and a division be equivalent?
- What's the inverse relationship?
- What does commutative mean?
- How do you check on the balance?

Exit: Each child writes ONE equivalent pair on a sticky note: ' $___ \times ___ = ___ \div ___$ '.

Lesson 4: Many Ways to Make 100, 250, 500

Goal: There are MANY ways to make any whole number up to 1000. Mix all four operations.

Expectation: C2.3

Learning Goal: "I can find many equivalent expressions for whole numbers up to 1000."

Materials: M2_balance_mat, M5_target_cards, M7_two_colour_counters, M8_number_line

1. Minds On (12 mins): Eddy's Many-Ways Hunt

Eddy: How many ways to make 100? Vex: SO many. Let's find them all. Eddy opens: 'MORE WAYS, MORE OPERATIONS, MORE math!'

Bring children to the carpet. Eddy + Vex puppets. 'Friends — yesterday we tested pairs. Today we hunt MANY ways to make a number.'

2. Action (25 mins): Many-Ways Stations

- Whole group: find 6+ equivalents for 200.
- Three stations.
- Stretch challenge at Station C: include a division and a 3-term expression.
- Discuss patterns.
- Circulate to record formative evidence on the equivalent-relationships tracker.
- STRATEGY COMPARE: Two pairs share their 5-expression paths to 250.

3. Consolidation (13 mins): Eddy's Many-Ways Brag

Discuss:

- How many ways to make 100?
- Can a division equal a multiplication for the same target?
- What's a 3-term equivalent expression for 250?
- Why are there so many ways?

Exit: Eddy's many-ways refrain — class chants: 'Eddy says: MORE WAYS, MORE OPERATIONS, MORE math!' (3x).

Lesson 5: Balance Puzzle Capstone

Goal: We can create balance puzzles using variables, equivalent expressions, and big numbers.

Expectation: C2.1, C2.2, C2.3

Learning Goal: "I can create and solve a balance puzzle using a variable."

Materials: M2_balance_mat, M5_target_cards, M6_capstone_template, M7_two_colour_counters

1. Minds On (12 mins): Eddy's Final Balance Puzzles

Now you create your own balance puzzles. Pick a target. Add a variable. Use 4 operations.

Bring children to the carpet. Eddy + Vex together. 'Friends — today you become EQUATION ENGINEERS. Create a balance puzzle that uses a VARIABLE, EQUIVALENT EXPRESSIONS, and a TARGET number.'

2. Action (25 mins): Balance Puzzle Capstone

- Hand out capstone templates.
- Children solve their puzzle and write 2 equivalent expressions for the target.
- Stretch challenge: include all 4 operations in the equivalents.
- Gallery walk: children swap and solve each other's puzzles.
- Whole-class share: 5-6 children share aloud.

3. Consolidation (13 mins): Eddy's Final Brag and Certificate Ceremony

Discuss:

- What's the most surprising puzzle you saw today?
- What does the variable stand for in your puzzle?
- What's an equivalent for your target?
- What's something in your life you'd model with a balance puzzle?

Exit: Each child receives a Junior Equation Engineer certificate.

Name: _____

Date: _____

Eddy's Variables

Learning Goal: I can describe how variables are used in real-life contexts.

Mission

1. Identify the variable in 6 real-life examples.
2. Sort 6 variables into UNKNOWN (one fixed value we need to find) or VARYING (changes over time).
3. Write 3 variables from your own life.

Part 2: Unknown or Varying

Sort 6 variables into UNKNOWN (one fixed value we need to find) or VARYING (changes over time).

[illegible]

Date: _____

Learning Goal: I can determine whether addition and subtraction expressions are equivalent.

1. Test 8 pairs of add/sub expressions.
2. For each given expression, write a different expression that's equivalent.
3. Write 3 of your own equivalent pairs using both + and -.

Write 3 of your own equivalent pairs using both + and -.

[illegible]

Name: _____

Date: _____

Eddy's Mult/Div Test

Learning Goal: I can determine whether multiplication and division expressions are equivalent.

Mission

1. Test 8 pairs of mult/div expressions.
2. Find an equivalent expression using a DIFFERENT operation.
3. Write 3 cross-operation equivalent pairs (multiplication = division).

Part 3: My Cross-Op Pairs

Write 3 cross-operation equivalent pairs (multiplication = division).

Name: _____

Date: _____

Eddy's Many Ways

Learning Goal: I can find many equivalent expressions for whole numbers up to 1000.

Instructions for the Student:

1. List 8 different equivalent expressions for 100.
2. List 6 different equivalent expressions for 500.
3. Pick a target between 250 and 1000.

Part 3: My Target

Pick a target between 250 and 1000. List 5 different expressions.

Date: _____

Learning Goal: I can create and solve a balance puzzle using a variable.

1. Pick a target.
2. Solve the puzzle.
3. Write 2 equivalent expressions for your target.

Write 2 equivalent expressions for your target.

[illegible]

Variable in Real Life Chart

(M1_variable_chart)

Letter-size chart with everyday variable examples.

Equation Table

Variable	Stands for	Value

PRINT SPECS

Size: letter landscape

Colour: bw

Pages per set: 1

Laminate: yes

TEACHER PREP

1. Print 1 chart per pair.

2. Laminate.

3. Hand out Lesson 1.

4. Reuse later.

5. Reference.

USE IN CLASS

Quantity: 1 per pair plus 1 anchor

Lessons: 1

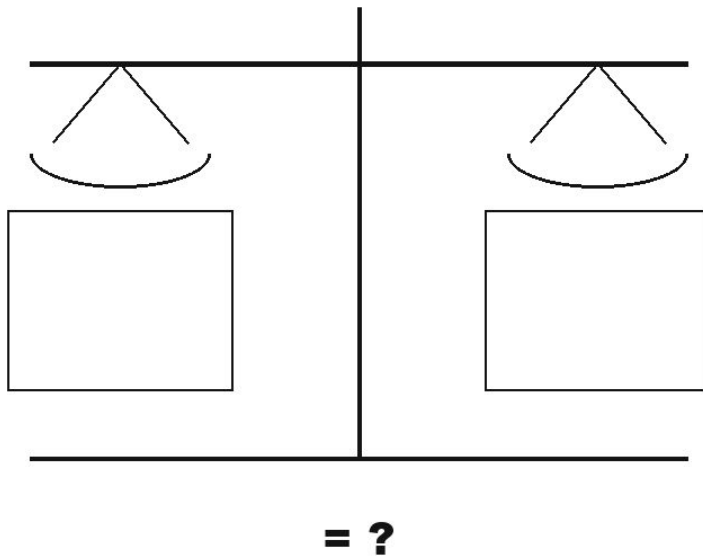
Prep: ~10 min

Big Balance Scale Mat

(M2_balance_mat)

Letter-size balance mat for numbers up to 1000.

Balance Mat



PRINT SPECS

Size: letter landscape
Colour: bw
Pages per set: 1
Laminate: yes

USE IN CLASS

Quantity: 1 per pair
Lessons: 2, 3, 4, 5
Prep: ~12 min

TEACHER PREP

1. Print 1 mat per pair.
2. Laminate.
3. Hand out Lesson 2.
4. Use dry-erase.
5. Reuse Lessons 3-5.

Addition/Subtraction

(M3_addition_cards)

24 cards for sorting equivalent vs not.

Addition Cards

$? + 3 = 8$	$5 + ? = 9$
$? + 4 = 12$	$10 = 6 + ?$
$? + 7 = 15$	$11 = ? + 4$
$? + 6 = 14$	$9 + ? = 13$

PRINT SPECS

Size: letter portrait

Colour: bw

Pages per set: 4

Laminate: yes

USE IN CLASS

Quantity: 24 cards

Lessons: 2

Prep: ~18 min

TEACHER PREP

1. Print 4 pages on cardstock.
2. Laminate.
3. Cut into 24 cards.
4. Sort by difficulty.
5. Distribute Lesson 2.

Multiplication/Division

(M4_multiplication_cards)

24 cards for sorting equivalent vs not.

Multiplication Cards

$? \times 3 = 12$	$4 \times ? = 20$
$? \times 5 = 25$	$6 \times ? = 24$
$? \times 8 = 32$	$? \times 7 = 21$
$9 \times ? = 36$	$? \times 4 = 28$

PRINT SPECS

Size: letter portrait

Colour: bw

Pages per set: 4

Laminate: yes

USE IN CLASS

Quantity: 24 cards

Lessons: 3

Prep: ~18 min

TEACHER PREP

1. Print 4 pages on cardstock.
2. Laminate.
3. Cut into 24 cards.
4. Sort by difficulty.
5. Distribute Lesson 3.

Equivalent-Target Cards

(M5_target_cards)

12 target cards for the equivalent-expression hunt.

Target Cards

10	20	50	100
5	25	75	200

PRINT SPECS

Size: letter portrait

Colour: bw

Pages per set: 2

Laminate: yes

TEACHER PREP

1. Print 2 pages on cardstock.

2. Laminate.

3. Cut into 12 cards.

4. Sort by size.

5. Distribute Lesson 4.

USE IN CLASS

Quantity: 12 cards

Lessons: 4

Prep: ~12 min

Balance Puzzle Capstone

(M6_capstone_template)

Letter-size capstone template.

Balance Puzzle Capstone	
1	Puzzle
2	Solution
3	Equivalent
4	Reflection

PRINT SPECS

Size: letter portrait

Colour: bw

Pages per set: 1

Laminate: no

TEACHER PREP

1. Print 1 per child.

2. Hand out Lesson 5.

3. Children fill in.

4. Collect for assessment.

5. Display.

USE IN CLASS

Quantity: 1 per child

Lessons: 5

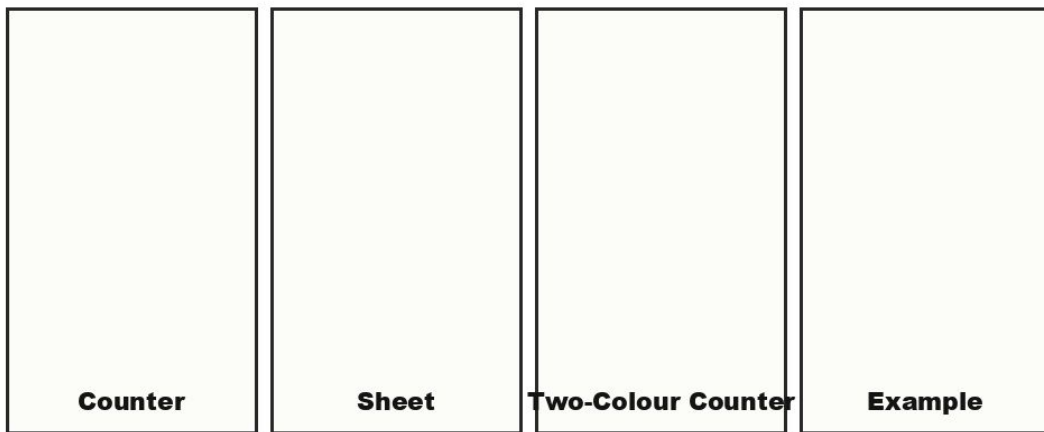
Prep: ~12 min

Two-Colour Counters

(M7_two_colour_counters)

30 counters per pair.

Two-Colour Counters



PRINT SPECS

Size: letter portrait

Colour: bw

Pages per set: 1

Laminate: yes

TEACHER PREP

1. Print 1 sheet per pair.

2. Laminate.

3. Cut into 30 counters.

4. Sort into pair bags.

5. Distribute Lessons 2-5.

USE IN CLASS

Quantity: 30 per pair

Lessons: 2, 3, 4, 5

Prep: ~15 min

Number Line to 1000

(M8_number_line)

Letter-size number line marked 0 to 1000 in 50s.

Number Line to 1000	
1	Major Ticks At Every 50 _____
2	Number _____
3	Number Line to 1000 _____
4	Example _____

PRINT SPECS

Size: letter landscape
Colour: bw
Pages per set: 1
Laminate: yes

TEACHER PREP

1. Print 1 line per pair.
2. Laminate.
3. Hand out Lesson 2.
4. Reuse Lessons 3-4.

USE IN CLASS

Quantity: 1 per pair
Lessons: 2, 3, 4
Prep: ~10 min

Equation Anchor Chart

(M9_equation_chart)

Anchor chart showing 4 operations and equivalence.

Equation Table

Equation	Variable	Solution

PRINT SPECS

Size: letter portrait

Colour: bw

Pages per set: 4

Laminate: yes

TEACHER PREP

1. Print 4 pages and tile.

2. Laminate.

3. Tape to wall Day 1.

4. Reference all week.

5. Roll for re-use.

USE IN CLASS

Quantity: 1 chart

Lessons: 1, 2, 3, 4

Prep: ~18 min

Eddy the Equation Engineer

(M10_engineer_poster)

Anchor poster of Eddy and Vex.

Engineer's Equations

- Both sides must **BALANCE**.
- What you do to one side, do to the other.
- Variables stand for numbers.
- Solve = isolate the variable.
- Check by substituting back.

PRINT SPECS

Size: letter portrait

Colour: bw

Pages per set: 4

Laminate: yes

USE IN CLASS

Quantity: 1 poster

Lessons: 1, 2, 3, 4, 5

Prep: ~18 min

TEACHER PREP

1. Print 4 pages and tile.

2. Laminate.

3. Tape to wall Day 1.

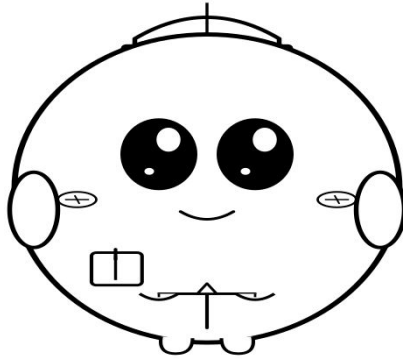
4. Reference all week.

5. Roll for re-use.

Eddy the Equation Engineer

(char_eddy_puppet)

Printable Eddy illustration on a popsicle stick.



Eddy the Engineer

PRINT SPECS

Size: letter portrait

Colour: bw

Pages per set: 1

Laminate: yes

TEACHER PREP

1. Print on cardstock.

2. Cut.

3. Glue popsicle stick.

4. Optional: laminate.

5. Store flat.

USE IN CLASS

Quantity: 1 puppet

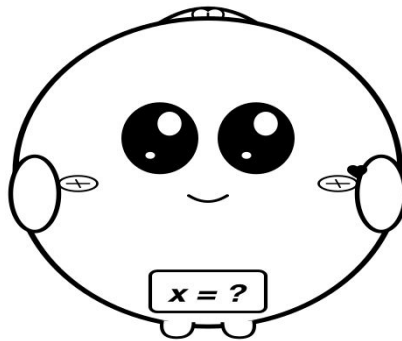
Lessons: 1, 2, 3, 4, 5

Prep: ~10 min

Vex the Variable Vet

(char_vex_puppet)

Printable Vex illustration on a popsicle stick.



Vex the Variable Vet

PRINT SPECS

Size: letter portrait

Colour: bw

Pages per set: 1

Laminate: yes

TEACHER PREP

1. Print on cardstock.

2. Cut.

3. Glue popsicle stick.

4. Optional: laminate.

5. Store flat.

USE IN CLASS

Quantity: 1 puppet

Lessons: 2, 3, 4, 5

Prep: ~10 min

Name: _____

Date: _____

Summative Assessment Rubric

Expectation	Level 1 (Beginning)	Level 2 (Developing)	Level 3 (Achieving)	Level 4 (Exceeding)
C2.1 describe how variables are used, and use them in various contexts as appropriate	Cannot describe a variable.	Describes variables with prompts.	Identifies variables and explains use.	Distinguishes unknown vs varying use.
C2.2 determine whether given pairs of addition, subtraction, multiplication, and division expressions are equivalent or not	Cannot test equivalence.	Tests equivalence with prompts and supports.	Tests equivalence across all 4 operations.	Cross-operation pairs; explains reasoning.
C2.3 identify and use equivalent relationships for whole numbers up to 1000, in various contexts	Cannot generate equivalents.	1-2 equivalents with prompts.	5+ equivalents using mixed operations.	Uses all 4 operations and 3+ term expressions.

Junior Equation Engineer Certificate

This certificate is proudly presented to:

For successfully completing the 5-Day Balanced Equations unit and demonstrating the skills of a Junior Equation Engineer. Eddy and Vex congratulate the student.

By completing this unit, this student has demonstrated:

- Describing variables
- Testing add/sub equivalence
- Testing mult/div equivalence
- Generating many equivalent expressions for big numbers
- Inventing and solving a balance puzzle

Marketplace Listing

A 5-day Ontario Grade-3 unit on Algebra (C2.1-C2.3) framed around 'Balanced Equations' — Eddy the Equation Engineer and Vex the Variable Vet.

Price: \$9.99 CAD

Pages: ~32

Classroom time: 250 min

Teacher prep: ~90 min

Top tags: grade-3, math, ontario-curriculum, algebra, variables, equivalent-expressions, canadian

What's included:

- 5 fully-scripted lesson plans with 50-minute durations and 250 minutes of total teaching time
 - 5 student worksheets aligned to each lesson
 - 1 mid-unit formative assessment (Equation Check-In)
 - 10 printable teacher manipulatives plus 2 character puppets: variable chart, big balance scale mat, addition/subtraction cards (24), multiplication/division...
- 1 end-of-unit reflection sheet with 5 self-assessment prompts
- 4 class observation trackers: 1 diagnostic + 3 formative
- 1 three-row summative rubric covering C2.1-C2.3 across 4 levels
- 1 summative-administration guide tying the rubric to Lesson 5 capstone
- Junior Equation Engineer certificate (one per child)
- Detailed expanded walkthroughs covering common misconceptions
- Differentiation tracks built into every lesson